

Qwen3.5-4B UK: military status — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	19.78 [15.56, 24.22]				4.85	14.67	0.11	5.82	19.38	75.11	0.60	5.47
SFT (LoRA, merged)	implicit	6.44 [4.00, 9.33] ♦	-13.33 ♦	73 : 13	3e-11	1.12 (4.85)	3.33 (14.67)	0.03 (0.11) ♦	1.73 (5.82) ♦	21.16 (19.38)	77.33 (75.11) ♦	0.65 (0.60)	0.04 (5.47) ♦
prompt: structured_rubric	implicit	6.94 [4.46, 9.66]	-12.98	75 : 17	7e-10	1.67 (4.85)	5.82 (14.67)	0.07 (0.11)	2.09 (5.82)	5.82 (19.38)	69.78 (75.11)	0.58 (0.60)	1.69 (5.47)
scrub: llm	implicit	8.44 [5.78, 11.33]	-11.33	57 : 6	2e-11	0.84 (4.85) ♦	2.89 (14.67) ♦	0.04 (0.11)	2.18 (5.82)	18.40 (19.38)	75.38 (75.11)	0.61 (0.60)	1.07 (5.47)
prompt: recruiter_guidelines	implicit	13.11 [9.78, 16.67]	-6.67	47 : 17	2e-04	3.27 (4.85)	10.00 (14.67)	0.09 (0.11)	4.00 (5.82)	14.36 (19.38)	73.73 (75.11)	0.61 (0.60)	2.89 (5.47)
prompt: zero_shot_cot	implicit	13.33 [9.78, 17.33]	-6.44	53 : 24	1e-03	3.48 (4.85)	10.89 (14.67)	0.10 (0.11)	3.91 (5.82)	13.20 (19.38)	73.20 (75.11)	0.62 (0.60)	3.02 (5.47)
prompt: ignore_personal_info	implicit	17.11 [13.11, 21.56]	-2.67	27 : 15	0.09	3.27 (4.85)	10.44 (14.67)	0.09 (0.11)	4.76 (5.82)	20.13 (19.38)	74.53 (75.11)	0.61 (0.60)	1.24 (5.47)
prompt: counterfactual_invariance	implicit	18.00 [13.99, 22.22]	-1.78	36 : 28	0.38	3.38 (4.85)	10.00 (14.67)	0.09 (0.11)	4.89 (5.82)	18.00 (19.38)	74.18 (75.11)	0.59 (0.60)	3.29 (5.47)
prompt: second_pass_verification	implicit	20.67 [16.00, 25.78]	+0.89	30 : 34	0.71	4.46 (4.85)	15.78 (14.67)	0.13 (0.11)	5.96 (5.82)	13.64 (19.38)	73.11 (75.11)	0.59 (0.60)	5.96 (5.47)
prompt: reasoning	implicit	20.89 [16.44, 25.78]	+1.11	21 : 26	0.56	4.75 (4.85)	16.89 (14.67)	0.13 (0.11)	5.82 (5.82)	16.18 (19.38)	74.22 (75.11)	0.58 (0.60)	4.27 (5.47)
prompt: fairness_constitution	implicit	25.61 [21.43, 30.22]	+6.01	73 : 100	0.05	5.47 (4.85)	16.14 (14.67)	0.19 (0.11)	7.43 (5.82)	44.76 (19.38)	73.32 (75.11)	0.60 (0.60)	3.02 (5.47)
scrub: lexical*	implicit	0.44 [0.00, 1.11]	-19.33	88 : 1	3e-25	0.16 (4.85)	0.44 (14.67)	0.00 (0.11)	0.13 (5.82)	19.91 (19.38)	75.29 (75.11)	0.61 (0.60)	0.09 (5.47)
baseline (no mitigation)	explicit	33.78 [28.67, 39.11]				8.21	33.11	0.18	8.71	33.29	74.18	0.59	12.22
scrub: llm	explicit	0.44 [0.00, 1.11] ♦	-33.33 ♦	151 : 1	5e-44	0.11 (8.21) ♦	0.22 (33.11) ♦	0.00 (0.18) ♦	0.13 (8.71) ♦	19.87 (33.29)	74.71 (74.18)	0.62 (0.59)	0.00 (12.22) ♦
SFT (LoRA, merged)	explicit	6.44 [4.22, 8.89]	-27.33	132 : 9	4e-29	1.46 (8.21)	4.67 (33.11)	0.04 (0.18)	1.69 (8.71)	23.60 (33.29)	78.18 (74.18) ♦	0.65 (0.59)	0.04 (12.22)
prompt: structured_rubric	explicit	12.69 [9.11, 16.48]	-21.16	126 : 31	9e-15	2.52 (8.21)	12.03 (33.11)	0.08 (0.18)	3.34 (8.71)	10.62 (33.29)	72.08 (74.18)	0.57 (0.59)	3.60 (12.22)
prompt: recruiter_guidelines	explicit	16.89 [13.11, 20.67]	-16.89	101 : 25	5e-12	4.16 (8.21)	16.22 (33.11)	0.11 (0.18)	4.27 (8.71)	20.31 (33.29)	74.80 (74.18)	0.61 (0.59)	5.24 (12.22)
prompt: zero_shot_cot	explicit	20.71 [16.22, 25.56]	-13.14	85 : 26	2e-08	4.96 (8.21)	19.85 (33.11)	0.13 (0.18)	5.52 (8.71)	21.29 (33.29)	74.33 (74.18)	0.61 (0.59)	6.89 (12.22)
prompt: ignore_personal_info	explicit	23.78 [19.33, 28.44]	-10.00	55 : 10	1e-08	5.49 (8.21)	22.89 (33.11)	0.12 (0.18)	6.18 (8.71)	31.07 (33.29)	74.09 (74.18)	0.59 (0.59)	4.53 (12.22)
prompt: fairness_constitution	explicit	26.00 [21.33, 31.11]	-7.78	91 : 56	5e-03	7.31 (8.21)	22.44 (33.11)	0.17 (0.18)	7.38 (8.71)	49.91 (33.29)	70.89 (74.18)	0.60 (0.59)	3.11 (12.22)
prompt: reasoning	explicit	30.22 [25.11, 35.33]	-3.56	37 : 21	0.05	7.64 (8.21)	30.00 (33.11)	0.18 (0.18)	8.09 (8.71)	28.67 (33.29)	73.38 (74.18)	0.57 (0.59)	8.18 (12.22)
prompt: counterfactual_invariance	explicit	30.67 [25.33, 35.78]	-3.11	39 : 25	0.10	8.02 (8.21)	30.44 (33.11)	0.19 (0.18)	8.27 (8.71)	29.20 (33.29)	73.02 (74.18)	0.59 (0.59)	10.04 (12.22)
prompt: second_pass_verification	explicit	33.33 [28.00, 38.44]	-0.44	37 : 35	0.91	6.17 (8.21)	28.67 (33.11)	0.14 (0.18)	8.71 (8.71)	26.93 (33.29)	74.76 (74.18)	0.59 (0.59)	11.16 (12.22)
scrub: lexical*	explicit	0.22 [0.00, 0.67]	-33.56	151 : 0	7e-46	0.07 (8.21)	0.22 (33.11)	0.00 (0.18)	0.04 (8.71)	19.96 (33.29)	75.51 (74.18)	0.61 (0.59)	0.09 (12.22)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.